Benjamin Johnston

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Technical Skills

Languages: C#, C, D, Python, JavaScript, TypeScript, Ruby, SQL
Test Automation: Selenium, WebDriver, W3C WebDriver, Regression Testing, Integration Testing, E2E Testing, Quality Gates, Unattended Execution
AI & Evaluation: LLM Evaluation, RAG Validation, Prompt Engineering, AI Safety, Red Teaming, LLM-as-a-Judge
Infrastructure: Docker, CI/CD, GitHub Actions, REST APIs, OAuth 2.0, JSON Schema

Experience

Software Engineer Jul 2021 – Present
Independent client engagements

- Designed and maintain 10+ cross-platform open-source libraries (700+ commits), including the only maintained W3C-compliant Selenium WebDriver client for D, with typed APIs, driver auto-detection for 4 browsers, and Page Object Model patterns.
- Engineered Warbler, an AI-assisted grading platform contracted for high-school use, with LLM subagent architecture, JSON schema enforcement, and hybrid deterministic + LLM grading (∼95% grading accuracy, halved instructor workload).

Automation Engineer (RPA & QA) Nov 2025 – Present
Handshake

- Eliminated 95% of manual effort across 400+ runtime hours by building AutoHandshake!, a Ruby/JavaScript Selenium RPA suite with unattended execution, UI-stall recovery, multi-tab orchestration, and self-test validation.
- Evaluated 4,000+ multimodal AI responses against structured rubrics (LLM-as-a-judge evaluation), earning the Handshake Model Validation 2 — Expert credential.
- Mentored contractors on test case interpretation and evaluation workflows, improving consistency and reducing rework from misaligned quality standards.

Test Engineer Dec 2025 – Jun 2026
Snorkel AI

- Established quality gates by building Docker-based reproducible validation sandboxes with CI/CD integration, catching defects pre-release and enforcing training-set integrity before pipeline ingestion.
- Designed adversarial test cases and boundary tests for Claude and ChatGPT across 2 contracted projects, applying strict schema constraints to surface model blindspots and reasoning failure modes.

Selected Projects

Akashi + Chemica | *RAG Pipeline, Python & D* | github.com/cetio/chemica

- Constructed a RAG pipeline aggregating compound data from 6 sources (PubChem, Wikipedia, PsychonautWiki, Erowid, PubMed, PMC) with 3D molecular visualization, structural similarity matching, and local embeddings.
- Extended the pipeline with LLM-powered source synthesis across 5 providers, enabling cross-source conflict resolution via local embeddings.

Intuit | *LLM SDK, D* | github.com/cetio/intuit

- Shipped the first native LLM SDK for D with 3 provider integrations (OpenAI, Claude, Qwen), OpenAI-compatible routing for OpenRouter and LiteLLM, and 10 example applications from RAG to multi-agent debates.
- Exposed native D functions as model tools via automatic JSON schema generation for type-safe tool use, with configurable-precision embeddings.

Selenium | *W3C WebDriver Client, D* | github.com/cetio/selenium

- Authored the only maintained W3C-compliant Selenium WebDriver client for D, published on DUB with 5 tagged releases, typed APIs, driver auto-detection for 4 browsers, and Page Object Model patterns.
- Shipped Selenium Grid support for distributed test execution; documented across 3 technical articles on state-driven automation and modular page object design.

Education & Certifications

Columbus State Community College Expected Dec 2026
AAS, Software Development
Handshake Model Validation 2 — Expert | MS Career Essentials Generative AI (NASBA) | SQL Programming (NASBA)